

The Open Generalized Mirror Theory Conjecture: Exact Multi-Graph Bounds Beyond the $O(2^{n/4})$ Regime

Theoretical Cryptography Note

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Abstract

Mirror Theory, initially introduced by Patarin (2003), provides lower bounds on the number of non-degenerate solutions to systems of bivariate linear equations over $\mathbb{F}_2^n$. While recent breakthroughs (e.g., Cogliati et al., 2023) have rigorously established the theory for component sizes up to $\xi_{\max} = O(2^{n/4}/\sqrt{n})$, the fully generalized conjecture for arbitrary dependency graphs up to the information-theoretic limit remains an open problem in theoretical cryptography. This document provides a self-contained statement of the open Generalized Mirror Theory Conjecture.

1 Formal Preliminaries

Let $V = \{p_1, p_2, \dots, p_q\}$ be a set of variables taking values in $\{0, 1\}^n$, where $N = 2^n$. We consider systems of bivariate affine equations over $\mathbb{F}_2^n$ defined on a multi-graph $G = (V, E)$.

Definition 1 (Affine Equation System). *Let $G = (V, E)$ be a connected graph with $q = |V|$ vertices and $m = |E|$ edges. Let $\lambda : E \rightarrow \{0, 1\}^n \setminus \{0^n\}$ be an edge-labeling function assigning non-zero target constants to each edge. The system of equations is given by:*

$$\mathcal{E}(G, \lambda) := \left\{ p_i \oplus p_j = \lambda_{i,j} \mid e = (p_i, p_j) \in E \right\}. \quad (1)$$

Definition 2 (Non-Degenerate Solutions). *A tuple $(P_1, \dots, P_q) \in (\{0, 1\}^n)^q$ is called a **valid (non-degenerate) solution** to $\mathcal{E}(G, \lambda)$ if:*

- i. $P_i \oplus P_j = \lambda_{i,j}$ for all $(p_i, p_j) \in E$.*
- ii. $P_i \neq P_j$ for all $i \neq j$ (all variables in V are pairwise distinct).*

We denote the set of all non-degenerate solutions by $\text{Sol}(\mathcal{E}(G, \lambda))$ and its cardinality by $h(G, \lambda) := |\text{Sol}(\mathcal{E}(G, \lambda))|$.

Definition 3 (Cycle Consistency and Component Size). *A system $\mathcal{E}(G, \lambda)$ is **cycle-consistent** if for every cycle C in G , the XOR sum of labels along the cycle evaluates to 0^n . Let $\xi_{\max}(G)$ denote the maximum size (number of vertices) of any connected component of G , and let $\nu(G) = |E| - |V| + c(G)$ be the circuit rank (cyclomatic number) of G , where $c(G)$ is the number of connected components.*

2 The Unconditional Generalized Mirror Theory Conjecture

While localized versions of Mirror Theory have been proven for trees and bounded-degree graphs up to $\xi_{\max} = O(N^{1/4})$, the full asymptotic statement for arbitrary graph topologies up to $O(N^{1/2})$ or higher query limits remains open.

Conjecture 1 (Generalized Mirror Theory Conjecture). *Let $G = (V, E)$ be a graph with $q = |V|$ vertices and $m = |E|$ edges. Let $\mathcal{E}(G, \lambda)$ be a cycle-consistent affine system. Assume that $q \leq o(N^{1/2})$ and that the maximum component size satisfies $\xi_{\max}(G) \leq o(N^{1/2})$.*

Then, the number of non-degenerate solutions $h(G, \lambda)$ satisfies the tight lower bound:

$$h(G, \lambda) \geq \frac{(N)_q}{N^m} \cdot \left(1 - O\left(\frac{q \cdot \xi_{\max}(G)}{N}\right)\right), \quad (2)$$

where $(N)_q = N(N-1)\dots(N-q+1)$ denotes the falling factorial.

Equivalently, for a collection of c connected tree components $T_1, \dots, T_c$ with $|V(T_k)| = q_k$, $q = \sum q_k$, and $m = q - c$:

$$h(G, \lambda) \geq \frac{N^q}{N^{q-c}} \prod_{i=0}^{q-1} \left(1 - \frac{i}{N}\right) (1 - \epsilon(q, \xi_{\max}, N)), \quad (3)$$

where $\epsilon(q, \xi_{\max}, N) \rightarrow 0$ as long as $q \cdot \xi_{\max}(G) = o(N)$.

3 State of the Art and Theoretical Gaps

Reference	Scope / Topology	Max Component ($\xi_{\max}$)	Status
Patarin (2003), Patarin (2005)	Linear Graphs / Trees	$\xi_{\max} = O(1)$	Gaps identified
Cogliati & Seurin (2018)	Circle Graphs ($\nu = 1$)	$\xi_{\max} = O(N^{1/3})$	Proven
Cogliati et al. (2023)	Arbitrary Forest Graphs	$\xi_{\max} = O(N^{1/4}/\sqrt{n})$	Fully Proven
Open Conjecture	Arbitrary Multi-Graphs	$\xi_{\max} = o(N^{1/2})$	OPEN

Table 1: Evolution of Mirror Theory bounds for system size q and component size $\xi_{\max}$.

Why the Conjecture Remains Open:

1. **High-Order Link Deletions:** Current algebraic techniques (such as link-deletion equations) rely on induction over component sizes. When $\xi_{\max}$ exceeds $O(N^{1/4})$, higher-order dependency intersections generate non-trivial error terms that cannot be absorbed without explicit structural cancellation lemmas.
2. **Multi-Cycle Dependencies:** For graphs with high cyclomatic numbers ($\nu(G) > 1$), handling non-equations ($P_i \neq P_j$) alongside linear constraints requires tracking exact character sums over non-abelian representations, which currently lack tight lower bounds.